AI PERSONAL STUDY ASSISTANT

FOR MACHINE LEARNING

Plan • Learn • Practise • Revise • Track Progress

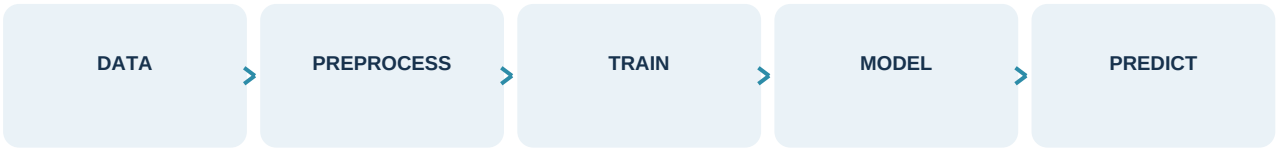
PREPARED BY AJMI.M

CS AIML BATCH

A Machine Learning focused study platform that combines an AI tutor, personalized planner, coding practice, notes, reminders and progress analytics in one website.

AI Personal Study Assistant for Machine Learning

The website helps ML students organize learning around concepts, mathematics, Python coding, datasets, algorithms, model training and evaluation. The AI can recommend what to learn next based on goals, deadlines, practice results and weak topics.



AI Tutor

Concept explanations, examples, code guidance and revision questions.

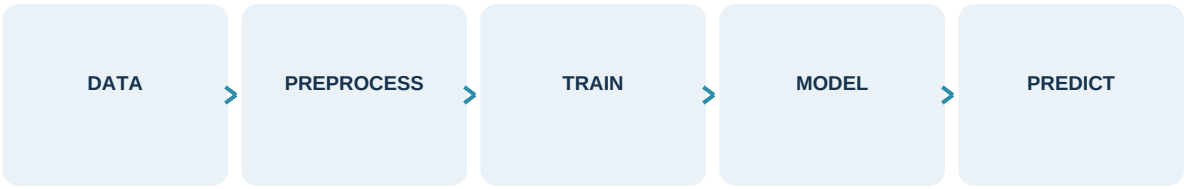
ML Planner

Schedules theory, coding, quizzes, projects and spaced revision.

Analytics

Tracks mastery, study time, quiz scores and weak areas.

Machine Learning Picture: From Data to Prediction

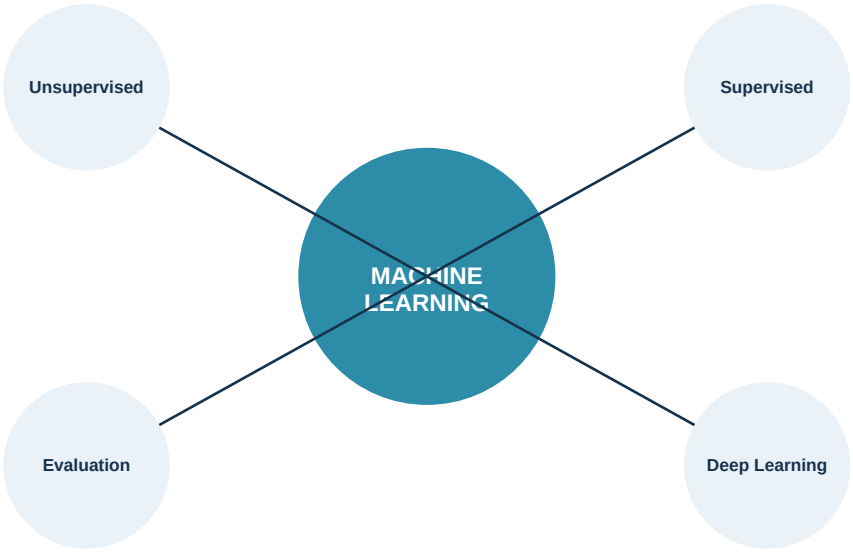


Students learn the same workflow through lessons, coding exercises, quizzes and projects.

Target Students

- AI/ML undergraduate students learning core Machine Learning subjects.
- Beginners who need simple explanations before moving to mathematical depth.
- Students preparing for university exams, coding interviews or ML certifications.
- Learners building projects who need a structured routine for theory and implementation.

ML Concept Map



Python
Learn + practise + revise

Regression
Learn + practise + revise

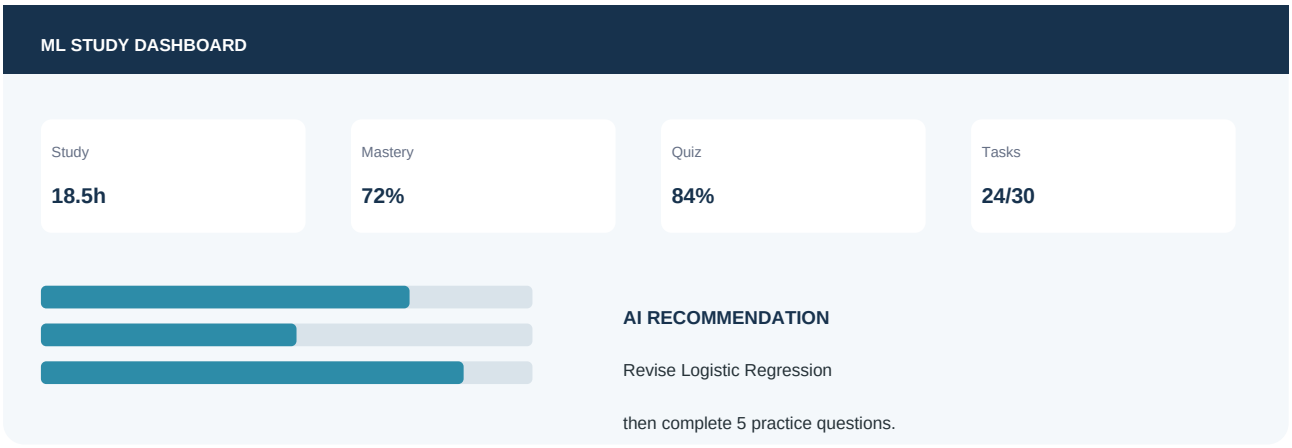
Classification
Learn + practise + revise

Clustering
Learn + practise + revise

Neural Networks
Learn + practise + revise

Model Evaluation
Learn + practise + revise

3. KEY WEBSITE PAGES + ML DASHBOARD VISUAL



Home / Landing
Project introduction and ML learning focus.

Onboarding
Skill level, subjects, goals, exam date and study hours.

Dashboard
Today's plan, deadlines, progress and AI recommendations.

ML Study Planner
Adaptive sessions for theory, coding and revision.

To-Do + Reminders
Assignments, projects, deadlines and alerts.

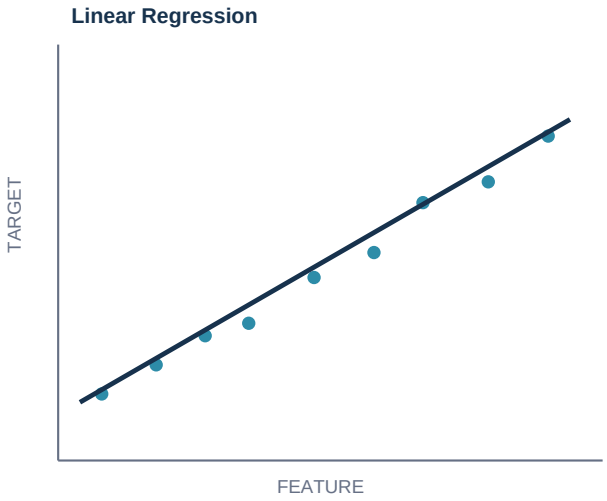
AI ML Chat
Ask, explain, practise, summarize and debug learning code.

Notes
Algorithms, formulas, code snippets and summaries.

Progress
Mastery, quiz scores, study time, streaks and weak topics.

Adaptive Planning Process

- Profile: select ML subjects, skill level, exam date and available hours.
- Goals: choose exam preparation, algorithm mastery, project completion or interview practice.
- AI Plan: allocate time across concepts, coding, quizzes, projects and revision.
- Adaptation: reschedule missed work and increase practice for weak topics.
- Daily Guidance: show the next best task and explain why it is recommended.



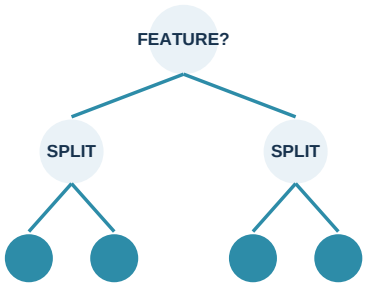
Example Session

15 min concept review 20 min Python implementation 10 min quiz 5 min reflection

AI Planning Rule

If quiz performance falls below the target, the next plan adds focused revision and extra practice instead of simply moving to a new topic.

Decision Tree



Clustering

To-Do List

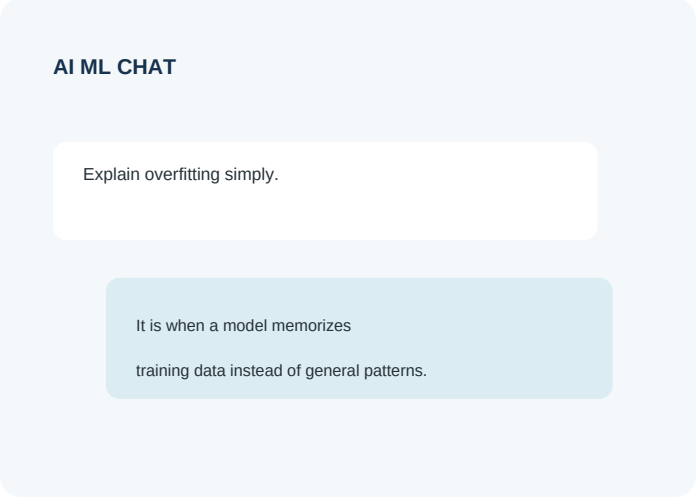
• Finish regression assignment • Implement decision tree • Practise clustering questions • Complete ML mini project • Review missed quiz questions

ML Notes

• Algorithm summaries • Formula sheets • Python code snippets • Dataset observations • AI-generated summaries • Search + topic tags

Smart Reminder Example

“Your weakest topic is Decision Trees. Tomorrow's 7:00 PM session includes a 15-minute concept review followed by two coding exercises.”



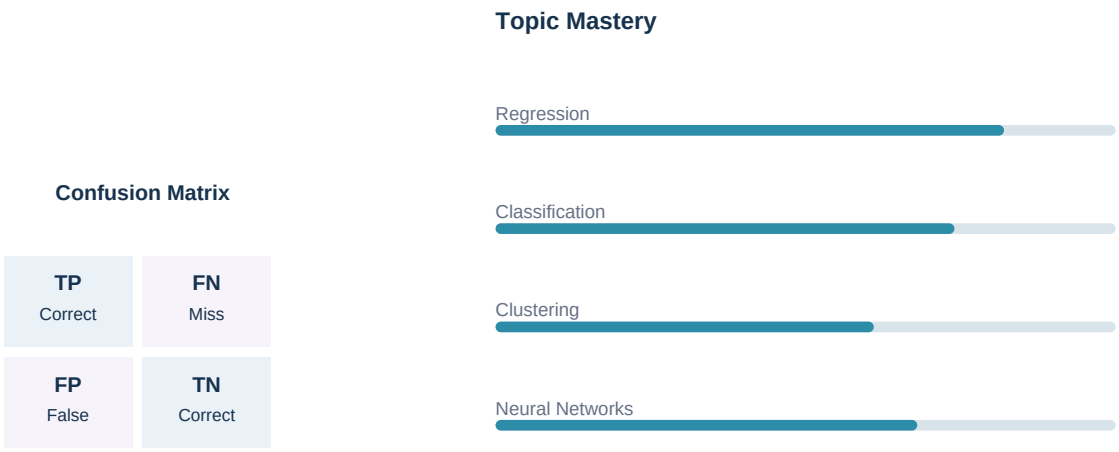
NEURAL NETWORK

AI Tutor Modes

- Concept mode — beginner-friendly explanations or deeper mathematical detail.
- Code mode — explain Python/ML code, errors and implementation decisions.
- Practice mode — generate MCQs, coding questions and interview questions.
- Revision mode — summaries, flashcards and quick tests.
- Project mode — break an ML project into manageable learning tasks.

Progress Dashboard

- Study hours and consistency over time.
- Topic mastery for regression, classification, clustering, neural networks and evaluation.
- Quiz performance and frequently missed concepts.
- Project completion and coding practice progress.
- AI-detected weak topics with recommended next actions.



Suggested Technology Stack

Frontend: HTML/CSS/JavaScript or React • Backend: Python Flask/FastAPI • ML: Python, NumPy, pandas, scikit-learn • Deep Learning: TensorFlow/PyTorch • Database: SQLite/PostgreSQL • AI: LLM/NLP service.

MVP: Dashboard + Planner + To-Do + AI ML Chat + Notes + Progress